1.60 Geometrical Configuration of Planes

Worked Solutions

1)

The three possible arrangements are: the planes are coincident; the planes are parallel; or the planes intersect in a line.

2)

The eight arrangements are:

1. Π_1, Π_2, Π_3 intersect at one point (one solution).
2. All three planes are coincident (infinitely many solutions).
3. Π_1 and Π_2 are coincident and Π_3 is parallel (no solutions).
4. Π_1 and Π_2 are coincident and Π_3 intersects them in a line (infinitely many solutions).
5. All three planes are parallel (no solutions).
6. Π_1 and Π_2 are parallel and Π_3 intersects them in different lines (no solutions).
7. The planes form a triangular prism (no solutions).
8. The planes form a sheaf: they all intersect in a line (infinitely many solutions).

3)

The three possibilities are: the planes intersect at one point; form a triangular prism; or form a sheaf, intersecting in a line.

4)

The planes form a sheaf: they intersect in a line, so there are infinitely many solutions.

5 a)

$$A^{-1} = \begin{pmatrix} -\frac{8}{11} & \frac{13}{2} & -\frac{6}{13} \\ \frac{11}{2} & -\frac{17}{2} & 4 \\ \frac{13}{2} & -\frac{21}{2} & 5 \end{pmatrix}.$$

5 b)

Using the inverse matrix from (a),

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 10 \\ 12 \\ 8 \end{pmatrix} = \begin{pmatrix} 28 \\ -15 \\ -21 \end{pmatrix}.$$

Thus $x = 28$, $y = -15$ and $z = -21$.

5 c)

The three planes intersect at the unique point $(28, -15, -21)$.

6)

$$\begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 1 \\ 4 & -1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Using the inverse matrix to solve gives $(x, y, z) = (1, 0, 1)$. So the three planes intersect at a unique point, $(1, 0, 1)$.

7)

$$\begin{pmatrix} 1 & 4 & -2 \\ -6 & -7 & 1 \\ 2 & 5 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 15 \\ -40 \\ 22 \end{pmatrix}.$$

The inverse matrix is $\begin{pmatrix} 9 & -2 & -10 \\ -10 & 2 & 11 \\ -16 & 3 & 17 \end{pmatrix}$, so

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ 12 \\ 14 \end{pmatrix}.$$

Thus the planes meet at the unique point $(-5, 12, 14)$.

8)

$$\begin{vmatrix} 4 & 1 & 9 \\ 3 & 2 & -5 \\ 4 & 3 & -10 \end{vmatrix} = -1 \neq 0.$$

As the determinant is nonzero, the three planes always meet at a unique point, irrespective of the value of k.

9 a)

Π_1 can be written as $3x + y - 2z = \frac{1}{2}$, so it is parallel to Π_3. Π_2 is not parallel to Π_1 or Π_3. Hence Π_1 and Π_3 are parallel, while Π_2 intersects each in a different line; there are no solutions.

9 b)

Π_2 can be written as $3x - y + 5z = 4$, which is the same equation as Π_3. Π_1 is not parallel to Π_2 or Π_3. Hence Π_2 and Π_3 are coincident, while Π_1 intersects them in a line (infinitely many solutions).

9 c)

Multiplying the equations for Π_1, Π_2 and Π_3 by $\frac{1}{5}$, 3 and $\frac{2}{7}$ respectively gives, in each case, $4x + 6y - 3z = -\frac{6}{5}$. Hence all three planes are coincident (infinitely many solutions).

10)

None of the planes are parallel or coincident. The determinant of the coefficient matrix is

$$\begin{vmatrix} 11 & -3 & 1 \\ 1 & 0 & 2 \\ 3 & -1 & -1 \end{vmatrix} = 0,$$

so the planes do not meet at a unique point. Let $z = 0$:

$$11x - 3y = 2 \quad (1), \qquad x = -5 \quad (2), \qquad 3x - y = 4 \quad (3).$$

Solving (1) and (2) gives $x = -5$, $y = -19$. Checking in (3): $3(-5) - (-19) = 4$. The equations are consistent, so the planes form a sheaf (infinitely many solutions).

11)

None of the planes are parallel or coincident. The determinant of the coefficient matrix is zero, so they do not meet at a unique point. Let $z = 0$:

$$4x - 21y = 10 \quad (1), \qquad 8x + 2y = 3 \quad (2), \qquad 10x + 8y = 7 \quad (3).$$

Solving (1) and (2) gives $x = \frac{83}{176}$ and $y = -\frac{17}{44}$. Checking in (3):

$$10\left(\frac{83}{176}\right) + 8\left(-\frac{17}{44}\right) = \frac{13}{8} \neq 7.$$

The equations are inconsistent, so the planes form a triangular prism (no solutions).

12)

None of the planes are parallel or coincident. The determinant of the coefficient matrix is zero, so they do not meet at a unique point. Let $z = 0$:

$$5x + 6y = 6 \quad (1), \qquad 2x - 3y = 6 \quad (2), \qquad x - 3y = 12 \quad (3).$$

Solving (1) and (2) gives $x = 2$ and $y = -\frac{2}{3}$. Checking in (3): $2 - 3(-\frac{2}{3}) = 4 \neq 12$. The equations are inconsistent, so the planes form a triangular prism (no solutions).

13 a)

None of the planes are parallel. When $k = 0$, $(0, 0, 0)$ is a solution, so all three planes pass through the origin. Hence they intersect at the unique point $(0, 0, 0)$.

13 b)

The determinant of the coefficient matrix is zero, so the planes do not meet at a unique point. Let $z = 0$:

$$2x + 4y = k \quad (1), \qquad x + 19y = 0 \quad (2), \qquad -2x + 4y = 0 \quad (3).$$

Solving (2) and (3) gives $x = 0$, $y = 0$. Substitution into (1) gives $0 = k$, which is false when $k \neq 0$. Thus the equations are inconsistent and the planes form a triangular prism (no solutions).

14 a)

When $k = 1$,

$$\Pi_1 : 2x + y + z = 1, \quad \Pi_2 : x + 3y + z = 1, \quad \Pi_3 : 4x + 2y + 2z = 1.$$

Π_1 and Π_3 are parallel because Π_3 can be written as $2x + y + z = \frac{1}{2}$, by multiplying through by $\frac{1}{2}$. Thus Π_1 and Π_3 are parallel, while Π_2 intersects them in different lines (no solutions).

14 b)

When $k = 3$,

$$\Pi_1 : 2x + 3y + 3z = 3, \quad \Pi_2 : 3x + 3y + 3z = 3, \quad \Pi_3 : 4x + 2y + 2z = 3.$$

None of the planes are parallel or coincident, and

$$\begin{vmatrix} 2 & 3 & 3 \\ 3 & 3 & 3 \\ 4 & 2 & 2 \end{vmatrix} = 0,$$

so the planes do not meet at a unique point. Let $z = 0$:

$$2x + 3y = 3 \ (1), \quad 3x + 3y = 3 \ (2), \quad 4x + 2y = 3 \ (3).$$

Solving (1) and (2) gives $x = 0$, $y = 1$. Checking in (3), $4(0) + 2(1) = 2 \neq 3$. The equations are inconsistent, so the planes form a triangular prism (no solutions).

15 a)

Since the planes do not meet at a unique point,

$$\begin{vmatrix} 2 & 4 & 5 \\ 3 & k & 4 \\ -2 & 4 & k \end{vmatrix} = 0.$$

Expanding,

$$2(k^2 - 16) - 4(3k + 8) + 5(12 + 2k) = 0,$$

$$2k^2 - 32 - 12k - 32 + 60 + 10k = 0,$$

$$2k^2 - 2k - 4 = 0 \Rightarrow k^2 - k - 2 = 0 \Rightarrow (k - 2)(k + 1) = 0.$$

Thus $k = 2$ or $k = -1$.

15 b)

When $k = 2$, the planes are $2x + 4y + 5z = 23$, $3x + 2y + 4z = 20$, $-2x + 4y + 2z = 6$. None are parallel or coincident. Set $z = 0$:

$$2x + 4y = 23 \ (1), \quad 3x + 2y = 20 \ (2), \quad -2x + 4y = 6 \ (3).$$

Solving (1) and (2) gives $x = \frac{17}{4}$ and $y = \frac{29}{8}$. Substitution in (3) gives $-2(\frac{17}{4}) + 4(\frac{29}{8}) = 6$. The equations are consistent, so the planes form a sheaf (infinitely many solutions). When $k = -1$, the planes are $2x + 4y + 5z = 23$, $3x - y + 4z = 20$, $-2x + 4y - z = 6$. None are parallel or coincident. Set $z = 0$:

$$2x + 4y = 23 \ (1), \quad 3x - y = 20 \ (2), \quad -2x + 4y = 6 \ (3).$$

Solving (1) and (2) gives $x = \frac{103}{14}$ and $y = \frac{29}{14}$. Substitution in (3) gives $-2(\frac{103}{14}) + 4(\frac{29}{14}) = -\frac{45}{7} \neq 6$. The equations are inconsistent, so the planes form a triangular prism (no solutions).

16)

The determinant of the coefficient matrix is

$$\begin{vmatrix} k+1 & 3 & -2 \\ k & 4 & -\frac{1}{4}(k+8) \\ 0 & 4 & -1 \end{vmatrix} = (k+1)(-4+k+8) + 3k - 8k$$

$$= (k+1)(k+4) + 3k - 8k = k^2 + 5k + 4 + 3k - 8k = k^2 + 4 \geq 4.$$

Thus the determinant is nonzero for all real k, so the planes always meet at a unique point.

17 a)

The determinant of the coefficient matrix is

$$\begin{vmatrix} k & 0 & -1 \\ 1 & 3 & k \\ 1 & 2 & k \end{vmatrix} = k(3k-2k) - 1(2-3) = k^2 + 1 \geq 1.$$

Thus the planes always meet at a unique point.

17 b)

In matrix form,

$$\begin{pmatrix} k & 0 & -1 \\ 1 & 3 & k \\ 1 & 2 & k \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

The cofactor matrix is $\begin{pmatrix} k & -2 & 3 \\ 0 & k^2+1 & -k^2-1 \\ -1 & -2k & 3k \end{pmatrix}$, whose transpose is $\begin{pmatrix} k & 0 & -1 \\ -2 & k^2+1 & -2k \\ 3 & -k^2-1 & 3k \end{pmatrix}$. The determinant is k^2+1, so

$$A^{-1} = \frac{1}{k^2+1} \begin{pmatrix} k & 0 & -1 \\ -2 & k^2+1 & -2k \\ 3 & -k^2-1 & 3k \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{2}{k^2+1} \\ 1 \\ -\frac{2k}{k^2+1} \end{pmatrix}.$$

So the planes intersect at the unique point $(-\frac{2}{k^2+1}, 1, -\frac{2k}{k^2+1})$.

18 a)

Since the planes do not meet at a unique point,

$$\begin{vmatrix} 1 & k & -4 \\ -6 & 4 & -3 \\ k & 1 & 4 \end{vmatrix} = 0.$$

Expanding gives

$$19 - k(-24+3k) - 4(-6-4k) = 0,$$

$$19 + 24k - 3k^2 + 24 + 16k = 0 \Rightarrow 3k^2 - 40k - 43 = 0$$

$$\Rightarrow (3k-43)(k+1) = 0.$$

Thus $k = \frac{43}{3}$ or $k = -1$.

18 b)

When $k = \frac{43}{3}$, none of the planes are parallel or coincident. Set $z = 0$:

$$x + \frac{43}{3}y = -19 \ (1), \quad -6x + 4y = -42 \ (2), \quad \frac{43}{3}x + y = 19 \ (3).$$

Solving (1) and (2) gives $x = \frac{263}{45}$ and $y = -\frac{26}{15}$. Substitution into (3) gives $\frac{43}{3}(\frac{263}{45}) - \frac{26}{15} = \frac{2215}{27} \neq 19$. The equations are inconsistent, so the planes form a triangular prism (no solutions). When $k = -1$, Π_3 becomes $-x + y + 4z = 19$, which is the same equation as Π_1 after multiplying by -1. Thus Π_1 and Π_3 are coincident, while Π_2 intersects them in a line (infinitely many solutions).

19 a)

Since the planes do not meet at a unique point,

$$\begin{vmatrix} k & 2 & 3k \\ 4 & k & -6 \\ 1 & 0 & -6 \end{vmatrix} = 0.$$

Expansion gives $-6k^2 + 36 - 3k^2 = 0$, so $k^2 - 4 = 0$, hence $(k-2)(k+2) = 0$. Thus $k = 2$ or $k = -2$.

19 b)

When $k = 2$, setting $z = 0$ gives $2x + 2y = 2$ (1), $4x + 2y = -2$ (2), $x = 0$ (3). Solving (2) and (3) gives $x = 0$, $y = -1$. This fails (1), since $2(0) + 2(-1) = -2 \neq 2$. The equations are inconsistent, so the planes form a triangular prism (no solutions). When $k = -2$, setting $z = 0$ gives $-2x + 2y = 2$ (1), $4x - 2y = -2$ (2), $x = 0$ (3). Solving (2) and (3) gives $x = 0$, $y = 1$, which satisfies (1): $2(0) + 2(1) = 2$. Thus the equations are consistent and the planes form a sheaf (infinitely many solutions).

20 a)

The determinant must be zero:

$$\begin{vmatrix} -5 & 3 & k+3 \\ k & 2k & -4 \\ 2 & -9 & 1 \end{vmatrix} = 0.$$

$$-5(2k + 36) - 3(k + 8) + (k + 3)(-13k) = 0$$

$$-10k + 180 - 3k - 24 - 13k^2 - 39k = 0 \Rightarrow k^2 + 4k - 12 = 0$$

$$(k + 6)(k - 2) = 0, \quad k = -6 \text{ or } k = 2.$$

20 b)

When $k = -6$, none of the planes are parallel or coincident. Set $z = 0$:

$$-5x + 3y = 2 \text{ (1)}, \quad -6x - 12y = -5 \text{ (2)}, \quad 2x - 9y = 4 \text{ (3)}.$$

Solving (1) and (2) gives $x = -\frac{3}{26}$, $y = \frac{37}{78}$. Then $2(-\frac{3}{26}) - 9(\frac{37}{78}) = -\frac{9}{2} \neq 4$. The equations are inconsistent, so the planes form a triangular prism (no solutions). When $k = 2$, none of the planes are parallel or coincident. Set $z = 0$:

$$-5x + 3y = 2 \text{ (1)}, \quad 2x + 4y = -5 \text{ (2)}, \quad 2x - 9y = 4 \text{ (3)}.$$

Solving (1) and (2) gives $x = -\frac{23}{26}$, $y = -\frac{21}{26}$. Substitution into (3) gives $2(-\frac{23}{26}) - 9(-\frac{21}{26}) = \frac{11}{2} \neq 4$. Thus the equations are inconsistent and the planes form a triangular prism (no solutions).

21 a)

The determinant must be zero:

$$\begin{vmatrix} 4 & -2 & 3 \\ k+1 & 3 & k-2 \\ 4 & k-3 & 5 \end{vmatrix} = 0.$$

$$4(15 - (k^2 - 5k + 6)) + 2(5k + 5 - 4k + 8) + 3((k + 1)(k - 3) - 12) = 0$$

$$4(9 + 5k - k^2) + 2(k + 13) + 3(k^2 - 2k - 15) = 0$$

$$36 + 20k - 4k^2 + 2k + 26 + 3k^2 - 6k - 45 = 0 \Rightarrow k^2 - 16k - 17 = 0$$

$$(k - 17)(k + 1) = 0, \quad k = 17 \text{ or } k = -1.$$

21 b)

When $k = 17$, none are parallel or coincident. Set $z = 0$:

$$4x - 2y = 17 \ (1), \quad 18x + 3y = 15 \ (2), \quad 4x + 14y = -65 \ (3).$$

Solving (1),(2) gives $x = \frac{27}{16}$, $y = -\frac{41}{8}$. Substitution into (3) gives $4(\frac{27}{16}) + 14(-\frac{41}{8}) = -65$. Thus the equations are consistent and the planes form a sheaf (infinitely many solutions). When $k = -1$, none are parallel or coincident. Set $z = 0$:

$$4x - 2y = 17 \ (1), \quad 3y = 15 \ (2), \quad 4x - 4y = -65 \ (3).$$

Solving (1),(2) gives $x = \frac{27}{4}$, $y = 5$. Substitution into (3) gives $4(\frac{27}{4}) - 4(5) = 7 \neq -65$. Thus the equations are inconsistent and the planes form a triangular prism (no solutions).

22)

For the planes to form a sheaf, the determinant must be zero:

$$\begin{vmatrix} 2k & k+3 & k \\ 1 & -1 & 2 \\ 3 & 1 & -2 \end{vmatrix} = 0.$$

$$2k((-1)(-2) - 2(1)) - (k+3)(1(-2) - 2(3)) + k(1(1) - (-1)(3)) = 0$$

$$8k + 24 + 4k = 0 \Rightarrow k = -2.$$

When $k = -2$, the equations are $-4x + y - 2z = 3$ (1), $x - y + 2z = p$ (2), $3x + y - 2z = 1$ (3). None of the planes are parallel or coincident. For a sheaf, the equations must be consistent. Let $z = 0$; solving (1) and (3) gives $x = -\frac{2}{7}$, $y = \frac{13}{7}$. Substitute in (2):

$$-\frac{2}{7} - \frac{13}{7} = p \Rightarrow p = -\frac{15}{7}.$$

Thus $k = -2$ and $p = -\frac{15}{7}$.

23 a)

The determinant must vanish:

$$\begin{vmatrix} 6 & -1 & k \\ 2 & -3 & 5 \\ 4 & 2 & -3 \end{vmatrix} = 6(-1) + (-26) + 16k = 0 \Rightarrow 16k - 32 = 0 \Rightarrow k = 2.$$

23 b)

Compare coefficients in

$$m(6x - y) + n(2x - 3y) \equiv 4x + 2y :$$

$$6m + 2n = 4 \quad (1), \qquad -m - 3n = 2 \quad (2).$$

Solving (1) and (2) gives $m = 1$ and $n = -1$.

23 c)

Set $z = 0$ in the three plane equations to obtain $6x - y = 33$, $2x - 3y = 25$, and $4x + 2y = p$. Using the identity from (b),

$$p = 1(33) - 1(25) = 8.$$

24 a)

A^{-1} cannot exist if $\det A = 0$. Thus

$$3(10 \cdot 3 - 0(k-1)) + 4(1 \cdot 3 - 0 \cdot 2) + 17(1(k-1) - 10 \cdot 2) = 0,$$

$$90 + 12 + 17(k - 21) = 0 \Rightarrow 17k - 255 = 0 \Rightarrow k = 15.$$

Therefore A^{-1} exists if $k \neq 15$.

24 b)

Here $k = 14$. The matrix equation is

$$\begin{pmatrix} 3 & -4 & 17 \\ 1 & 10 & 0 \\ 2 & 13 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 17 \\ 0 \\ m \end{pmatrix}.$$

Its inverse is

$$A^{-1} = \begin{pmatrix} -\frac{30}{17} & -\frac{233}{17} & 10 \\ \frac{3}{17} & \frac{25}{17} & -1 \\ \frac{7}{17} & \frac{47}{17} & -2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 17 \\ 0 \\ m \end{pmatrix} = \begin{pmatrix} 10m - 30 \\ 3 - m \\ 7 - 2m \end{pmatrix}.$$

So the unique point of intersection is $(10m - 30, 3 - m, 7 - 2m)$.

24 c)

Here $k = 15$. None of the planes are parallel or coincident. Set $z = 0$:

$$3x - 4y = n \ (1), \quad x + 10y = 1 \ (2), \quad 2x + 14y = 2 \ (3).$$

Solving (2),(3) gives $x = 1$, $y = 0$. Substitution in (1) gives $3(1) - 4(0) = n$, so $n = 3$. Since the planes do not meet at a unique point and the equations are consistent, they form a sheaf (infinitely many solutions).

25 a)

Since the planes do not meet at a unique point,

$$\begin{vmatrix} 2 & -3 & k \\ 5 & -3 & k^2 \\ 4 & 12 & -1 \end{vmatrix} = 0.$$

$$2(3 - 12k^2) + 3(-5 - 4k^2) + 72k = 0$$

$$6 - 24k^2 - 15 - 12k^2 + 72k = 0 \Rightarrow 4k^2 - 8k + 1 = 0,$$

so $k = 1 \pm \frac{\sqrt{3}}{2}$.

25 b)

When $k = 1 + \frac{\sqrt{3}}{2}$, the plane equations are

$$\Pi_1 : 2x - 3y + \left(1 + \frac{\sqrt{3}}{2}\right) z = p, \quad \Pi_2 : 5x - 3y + \left(\frac{7}{4} + \sqrt{3}\right) z = 1, \quad \Pi_3 : 4x + 12y - z = 8.$$

None of the planes are parallel or coincident. If the equations were consistent, set $z = 0$:

$$2x - 3y = p \quad (1), \qquad 5x - 3y = 1 \quad (2), \qquad 4x + 12y = 8 \quad (3).$$

Solving (2) and (3) gives $x = \frac{1}{2}$ and $y = \frac{1}{2}$. Substituting into (1) gives $p = 2(\frac{1}{2}) - 3(\frac{1}{2}) = -\frac{1}{2}$. When $k = 1 - \frac{\sqrt{3}}{2}$, the plane equations are

$$\Pi_1 : 2x - 3y + \left(1 - \frac{\sqrt{3}}{2}\right) z = p, \quad \Pi_2 : 5x - 3y + \left(\frac{7}{4} - \sqrt{3}\right) z = 1 - \sqrt{3}, \quad \Pi_3 : 4x + 12y - z = 8.$$

None of the planes are parallel or coincident. If the equations were consistent, set $z = 0$:

$$2x - 3y = p \quad (1), \qquad 5x - 3y = 1 - \sqrt{3} \quad (2), \qquad 4x + 12y = 8 \quad (3).$$

Solving (2) and (3) gives $x = \frac{1}{2} - \frac{\sqrt{3}}{6}$ and $y = \frac{1}{2} + \frac{\sqrt{3}}{18}$. Substituting into (1) gives $p = 2(\frac{1}{2} - \frac{\sqrt{3}}{6}) - 3(\frac{1}{2} + \frac{\sqrt{3}}{18}) = -\frac{1}{2} - \frac{\sqrt{3}}{2}$.

25 c)

The planes do not meet at a unique point and the equations are inconsistent, so they form a triangular prism (no solutions). Thus either $k = 1 + \frac{\sqrt{3}}{2}$ and $p \neq -\frac{1}{2}$, or $k = 1 - \frac{\sqrt{3}}{2}$ and $p \neq -\frac{1}{2} - \frac{\sqrt{3}}{2}$.

26 a)

Multiplying the matrices gives

$$AB = \begin{pmatrix} 36 - 14k & 12k - 48 & k^2 - 15k + 44 \\ k - 4 & 12 - 8k & k^2 + 3k - 28 \\ k - 4 & 32 - 8k & k^2 + 2k - 44 \end{pmatrix}.$$

26 b)

When $k = 4$,

$$AB = \begin{pmatrix} -20 & 0 & 0 \\ 0 & -20 & 0 \\ 0 & 0 & -20 \end{pmatrix} = -20I.$$

Hence $A\left(-\frac{1}{20}B\right) = I$, so

$$A^{-1} = -\frac{1}{20}B = \begin{pmatrix} -\frac{1}{20} & \frac{2}{5} & -\frac{1}{4} \\ -\frac{3}{4} & 1 & -\frac{3}{4} \\ -\frac{7}{20} & \frac{4}{5} & -\frac{3}{4} \end{pmatrix}.$$

26 c)

In matrix form,

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 40 \\ 100 \\ -1000 \end{pmatrix}.$$

Using the inverse from (b),

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 40 \\ 100 \\ -1000 \end{pmatrix} = \begin{pmatrix} 288 \\ 820 \\ 816 \end{pmatrix}.$$

Therefore the planes intersect at the unique point $(288, 820, 816)$.

27 a)

A is singular when $\det A = 0$:

$$\begin{vmatrix} 7 & -1 & 2 \\ k & 4 & -2 \\ 1 & -6 & 3 \end{vmatrix} = 7(4 \cdot 3 - (-2)(-6)) + (3k + 2) + 2(-6k - 4) = 0.$$

Thus $-9k - 6 = 0$, giving $k = -\frac{2}{3}$.

27 b)

Let $X = \sqrt{x}$, $Y = \sqrt{y}$ and $Z = \sqrt{z}$. The equations can be written

$$\begin{pmatrix} 7 & -1 & 2 \\ -\frac{7}{9} & 4 & -2 \\ 1 & -6 & 3 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 49 \\ \frac{10}{3} \\ -6 \end{pmatrix}.$$

The inverse matrix is

$$\begin{pmatrix} 0 & -9 & -6 \\ \frac{1}{3} & 19 & \frac{112}{9} \\ \frac{2}{3} & 41 & \frac{245}{9} \end{pmatrix},$$

so

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 0 & -9 & -6 \\ \frac{1}{3} & 19 & \frac{112}{9} \\ \frac{2}{3} & 41 & \frac{245}{9} \end{pmatrix} \begin{pmatrix} 49 \\ \frac{10}{3} \\ -6 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \\ 6 \end{pmatrix}.$$

Therefore $\sqrt{x} = 6$, $\sqrt{y} = 5$ and $\sqrt{z} = 6$, giving $(x, y, z) = (36, 25, 36)$.

28 a)

The planes Π_1 and Π_2 share the point $(1, 2, 5)$. Their normal vectors are

$$\mathbf{n}_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \times \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \\ -5 \end{pmatrix},$$

$$\mathbf{n}_2 = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -5 \\ 2 \\ 11 \end{pmatrix}.$$

These are not multiples, so Π_1 and Π_2 are not coincident. To check whether $(1, 2, 5)$ lies on Π_3, solve

$$1 = 8 - 2p - q \quad (1), \qquad 2 = -3 + 4p - q \quad (2), \qquad 5 = 2 - 3p + 3q \quad (3).$$

Solving (1) and (2) gives $p = 2$ and $q = 3$, which also satisfies (3). Thus the three planes intersect at the unique point $(1, 2, 5)$.

28 b)

There are no obvious common points of intersection. The normal vectors of Π_1 and Π_2 are

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} \times \begin{pmatrix} 3 \\ 9 \\ 5 \end{pmatrix} = \begin{pmatrix} 43 \\ -11 \\ -6 \end{pmatrix},$$
$$\mathbf{n}_2 = \begin{pmatrix} 6 \\ 18 \\ 10 \end{pmatrix} \times \begin{pmatrix} 3 \\ 15 \\ -6 \end{pmatrix} = \begin{pmatrix} -258 \\ 66 \\ 36 \end{pmatrix} = -6 \begin{pmatrix} 43 \\ -11 \\ -6 \end{pmatrix}.$$

Thus Π_1 and Π_2 are parallel or coincident. The point $(3, 4, 1)$ lies on Π_1. Check whether it lies on Π_2 by solving

$$3 = 3 + 6m + 3n \quad (1), \qquad 4 = 10 + 18m + 15n \quad (2), \qquad 1 = -10 + 10m - 6n \quad (3).$$

Solving (1) and (2) gives $m = \frac{1}{2}$ and $n = -1$, which also satisfies (3). Hence $(3, 4, 1)$ lies on Π_2 and Π_1 and Π_2 are coincident. A normal vector to Π_3 is

$$\begin{pmatrix} 3 \\ 9 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ 10 \\ -4 \end{pmatrix} = \begin{pmatrix} -76 \\ 20 \\ 12 \end{pmatrix},$$

which is not a multiple of the other normal vectors. Therefore Π_3 intersects the coincident planes in a line.